Torque-Measuring Adapter For Spinal Surgical Instruments

Sruti Bapatla[1,2], Jacob Bryant[1,3], Jeffrey Chan[1,3], Anna Iacocca[1,3], Zoe Jacobs[1,4], Kol Rollins[1,2], Siya Scindia[1,4]

[1]Department of Biomedical Engineering, [2] Department of Mechanical Engineering, [3]Department of Chemical Engineering, [4] Department of Material Science & Engineering
Carnegie Mellon University, Pittsburgh, PA

Problem

- ~300,000 spinal fusion surgeries / year in the US[1] for scoliosis and other spinal pathologies
- No quantification of torque applied to screws
  - "Surgeon feel"
  - Inconsistent application of torque may potentially lead to complications after surgery

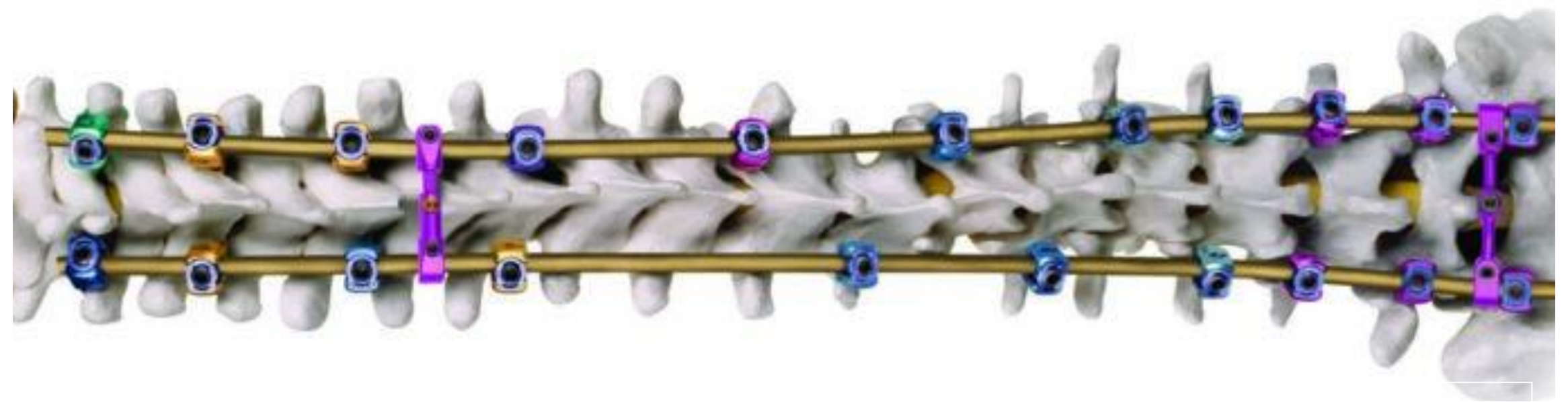


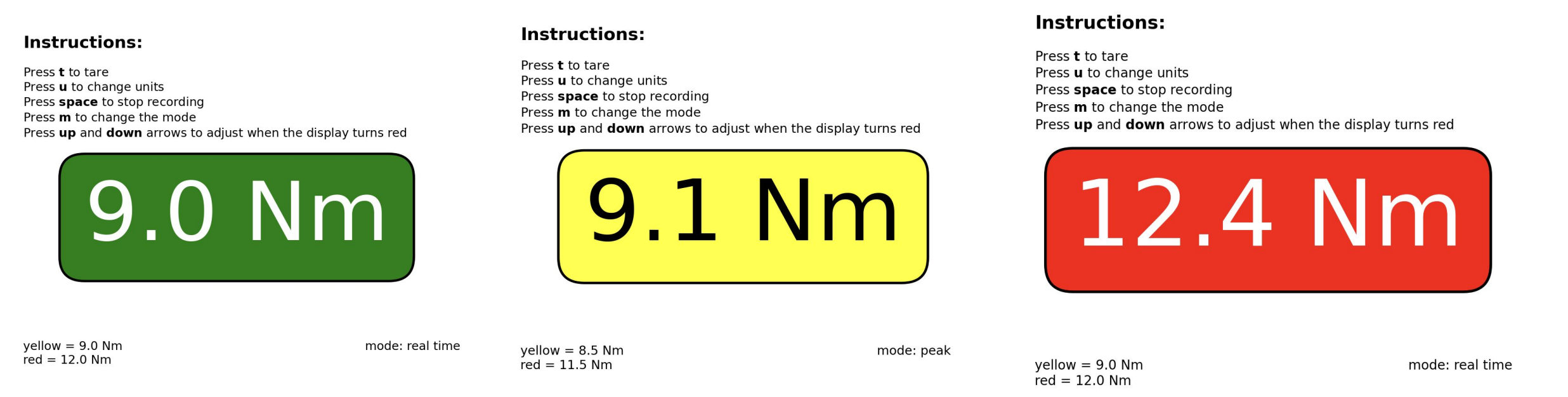
Figure 1: Spine surgically aligned with screws and rods

Proposed Solution

- Device that quantifies torque (up to 12.5 Nm) applied to screws during spinal surgery
- Compatible with Medtronic surgical drivers and handles
- Connected to OR computer monitors via wire

Device Key Features

- Measures, records, and displays torque
- Real time and peak display modes
- Tare option
- Customizable color feedback settings to indicate "safe" torque



Instructions:
Press **t** to tare
Press **u** to change units
Press **space** to stop recording
Press **m** to change the mode
Press **up** and **down** arrows to adjust when the display turns red

9.0 Nm

yellow = 9.0 Nm
red = 12.0 Nm
mode: real time

Instructions:
Press **t** to tare
Press **u** to change units
Press **space** to stop recording
Press **m** to change the mode
Press **up** and **down** arrows to adjust when the display turns red

9.1 Nm

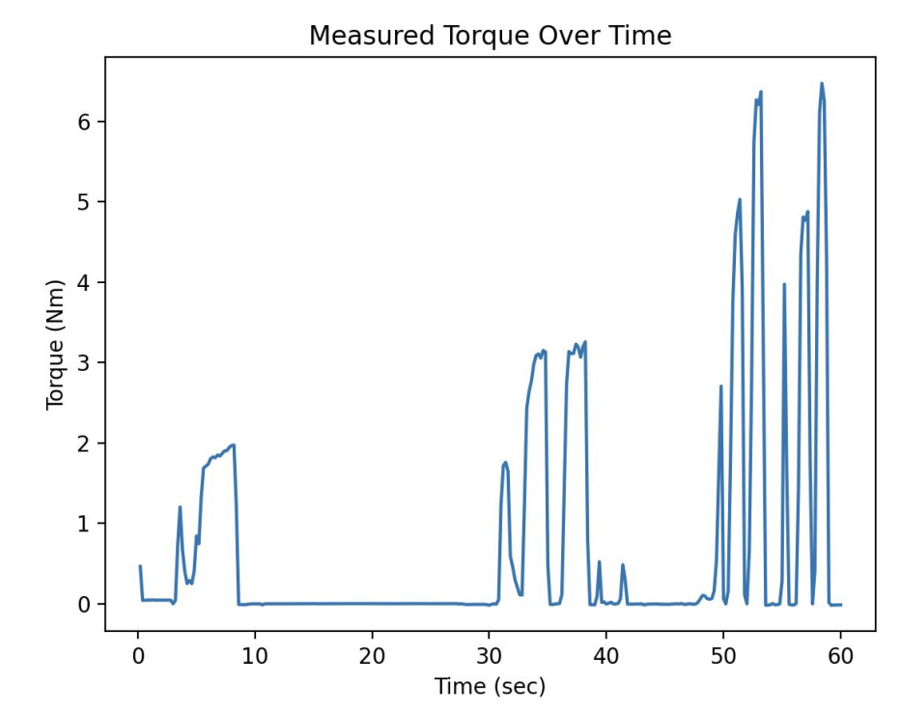
yellow = 8.5 Nm
red = 11.5 Nm
mode: peak

Instructions:
Press **t** to tare
Press **u** to change units
Press **space** to stop recording
Press **m** to change the mode
Press **up** and **down** arrows to adjust when the display turns red

12.4 Nm

yellow = 9.0 Nm
red = 12.0 Nm
mode: real time

Figure 2: Device computer-user interface showing color feedback



- Data plotted and saved as a .txt file when finished
- Cover for mechanical and electronic components to increase sterility

Figure 3: Torque vs. time data collected by device and displayed on computer-user interface

Complete Device Design

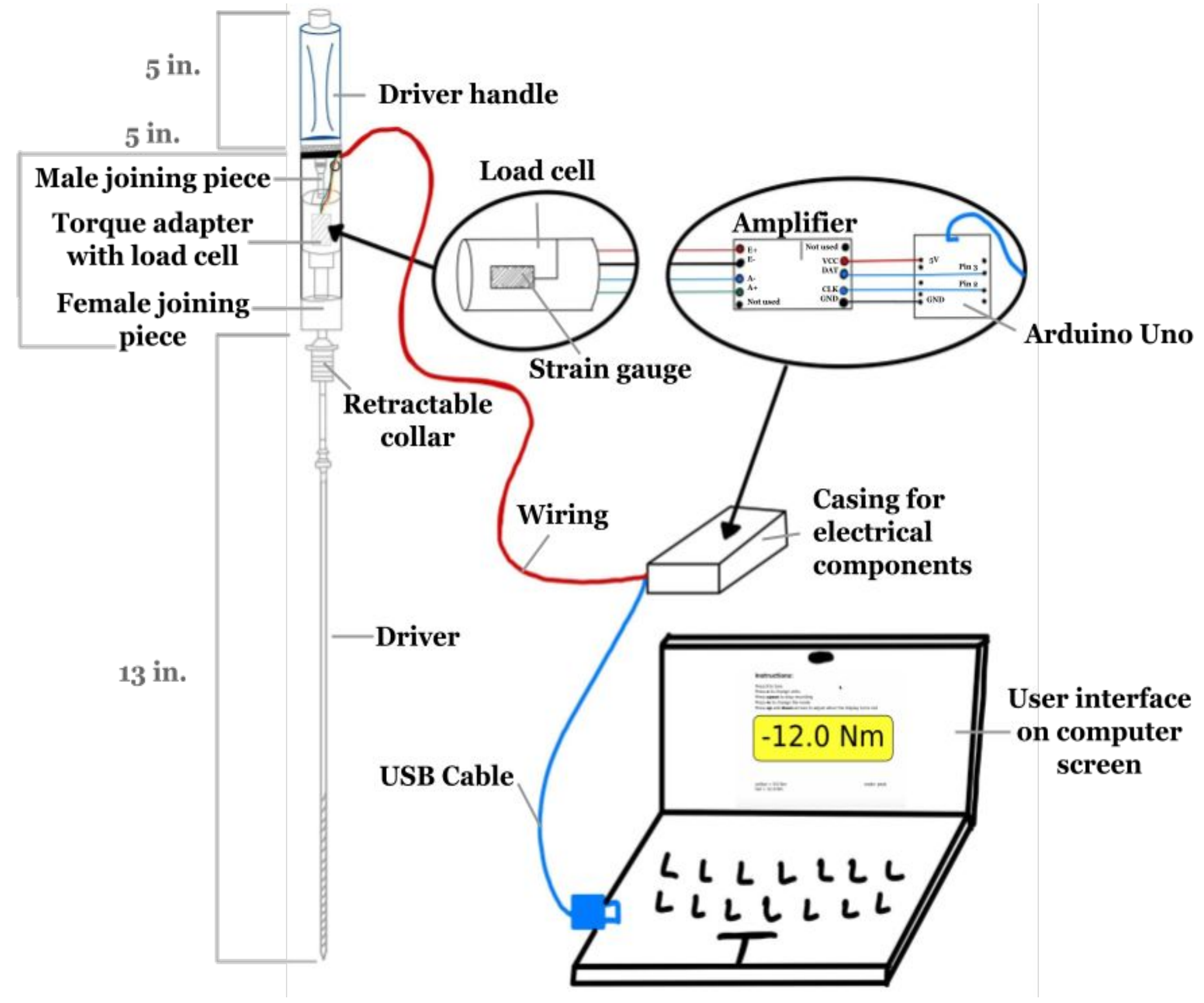


Figure 4. All device components (illustrated schematic, not drawn to scale).

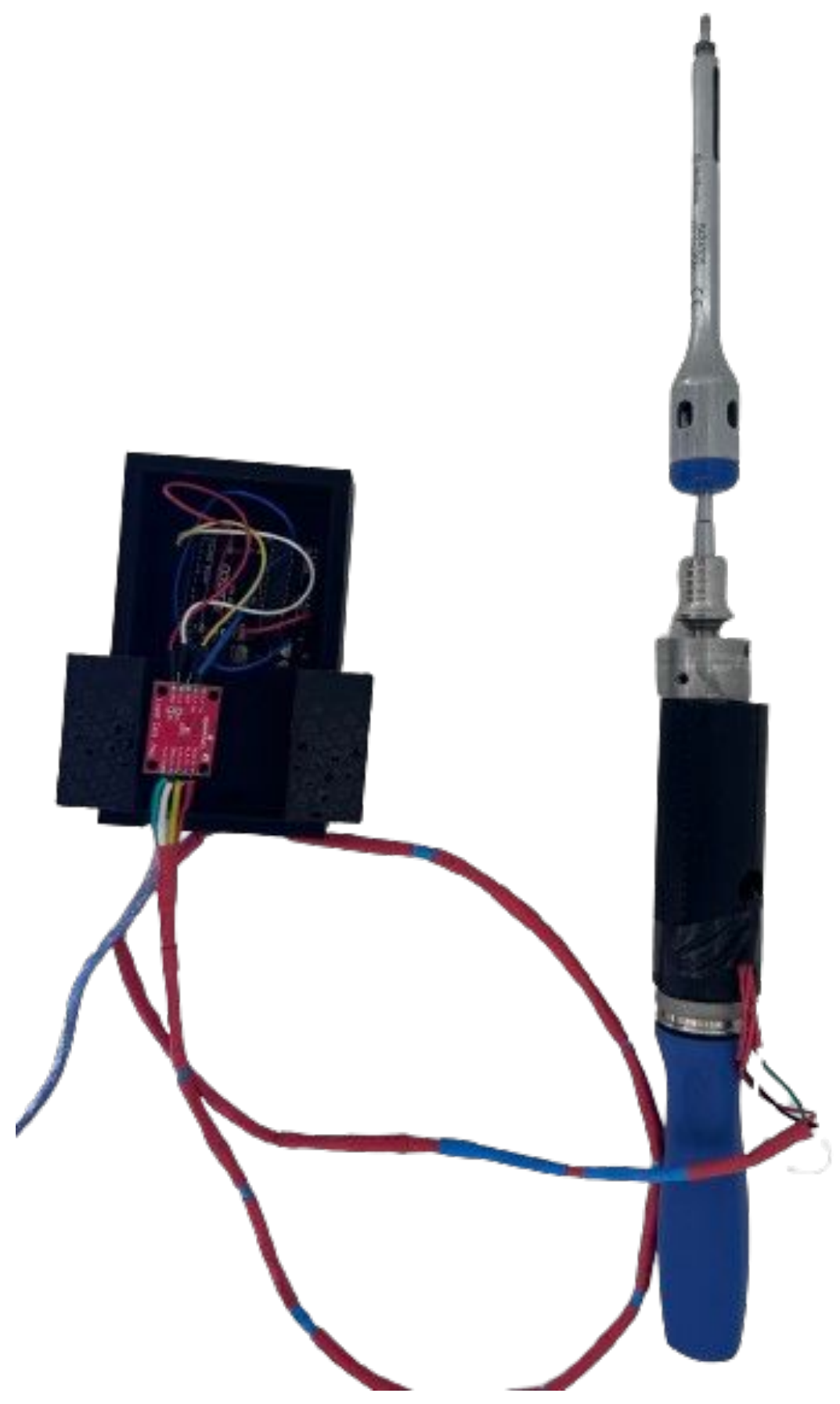


Figure 5. All device components (photographed).

Results

Calibration

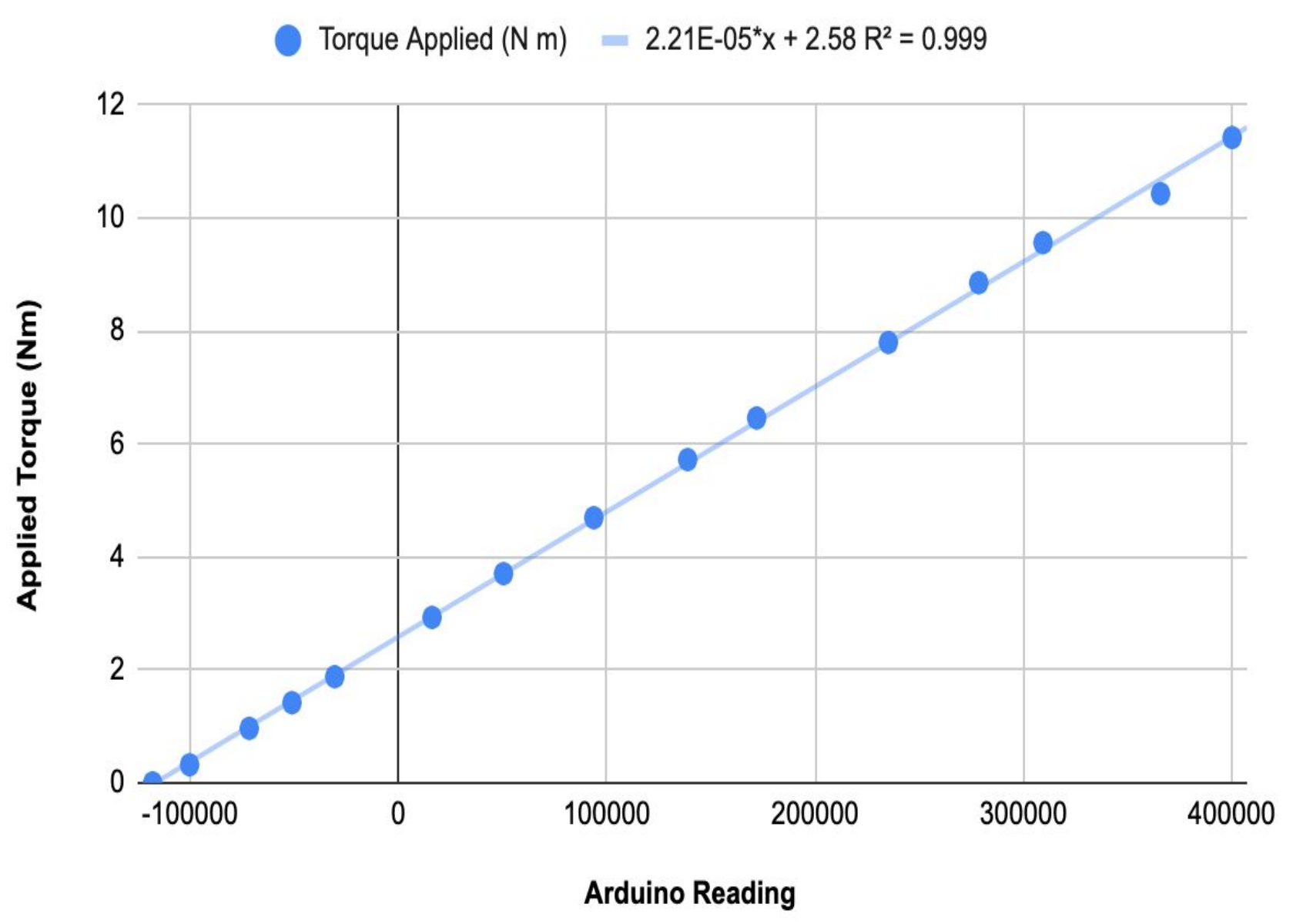


Figure 6: Sample calibration curve for torque meter using an applied torque wrench.

Testing

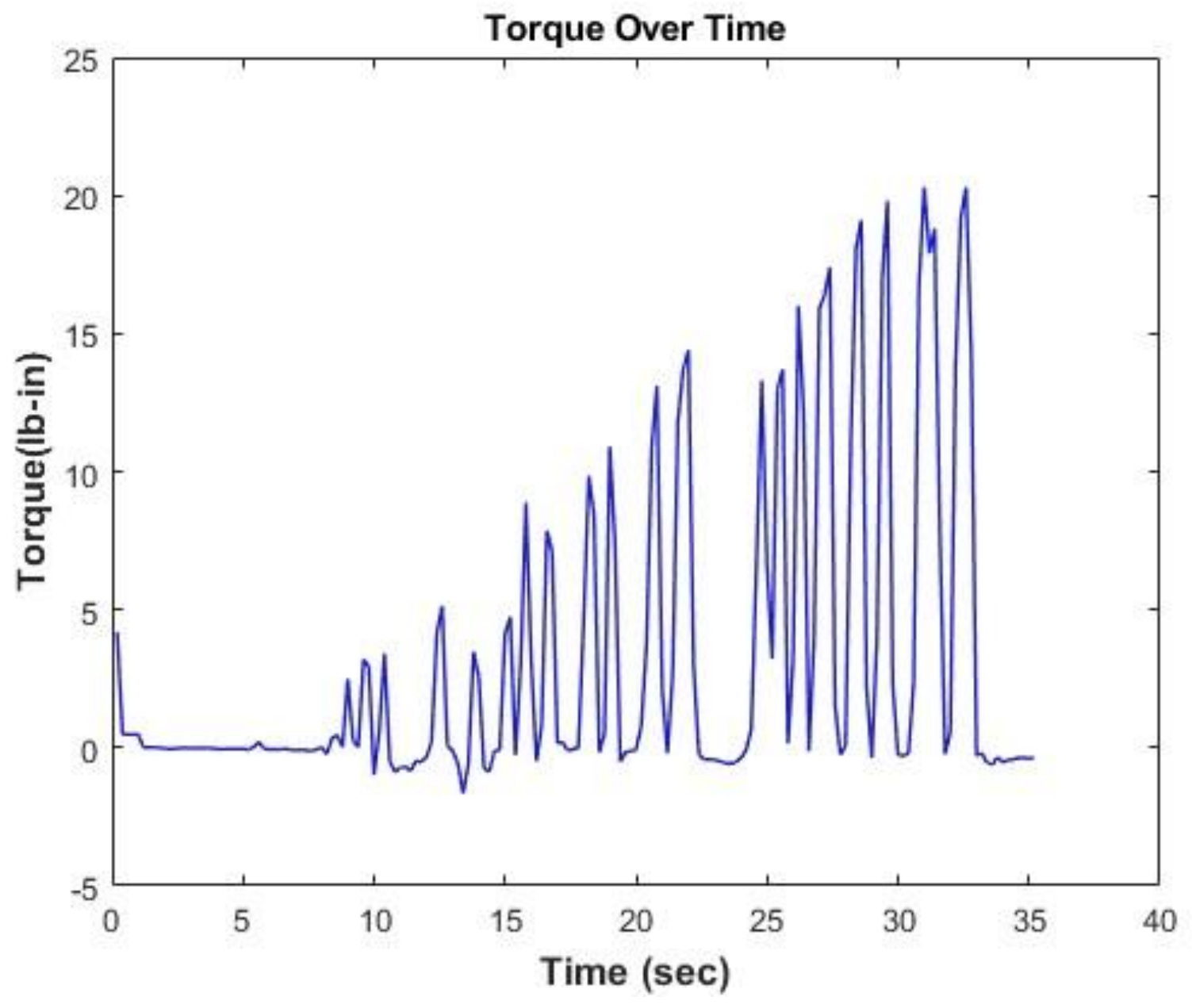


Figure 7: Torque measured by device while screwing pedicle screw into sawbone

- Calibration curve: linear relationship between Arduino output and applied torque
- Torque wrench upper limit = 10 N m, tolerance of .25 Nm
- Per ISO 6789:2017 [2, 3]

Stakeholder Testing

Strengths	Implemented Feedback
Provides live torque readings	Reduced length of adapter
Appropriate for cadaveric setting	Conversion between metric and US units
Graphical output of torque over time	Color feedback to indicate "safe" torque

Other Considerations

Medical Reimbursement

- Medicare/Medicaid: billing under spinal fusion procedures (subsets of the M43.2 code)

Intellectual Property

- No existing threat to patentability of device
  - Existing solutions include:
    - Torque wrenches/ gauges (not medical grade)[4, 5, 6]
    - Torque limiting devices (no real-time quantitative torque feedback)[7, 8]
    - Custom spinal screws with torque-reading sensors (still in early stages of development)[9]

Costs

- $100.29/unit (~$250.29 with labor)

Conclusion

- Can measure up to 11 Nm of Torque with a tolerance of .25 Nm
- Compatible with with Medtronic surgical drivers
- Can provide surgeons with quantitative feedback of torque with an easy-to-use computer-user interface

Future Work

- Make device wireless
- Manufacture out of stainless steel
- Develop water-tight seal for electronics
- Calibrate at higher torques
- Test device in cadaveric setting
- Develop computer app for UI

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